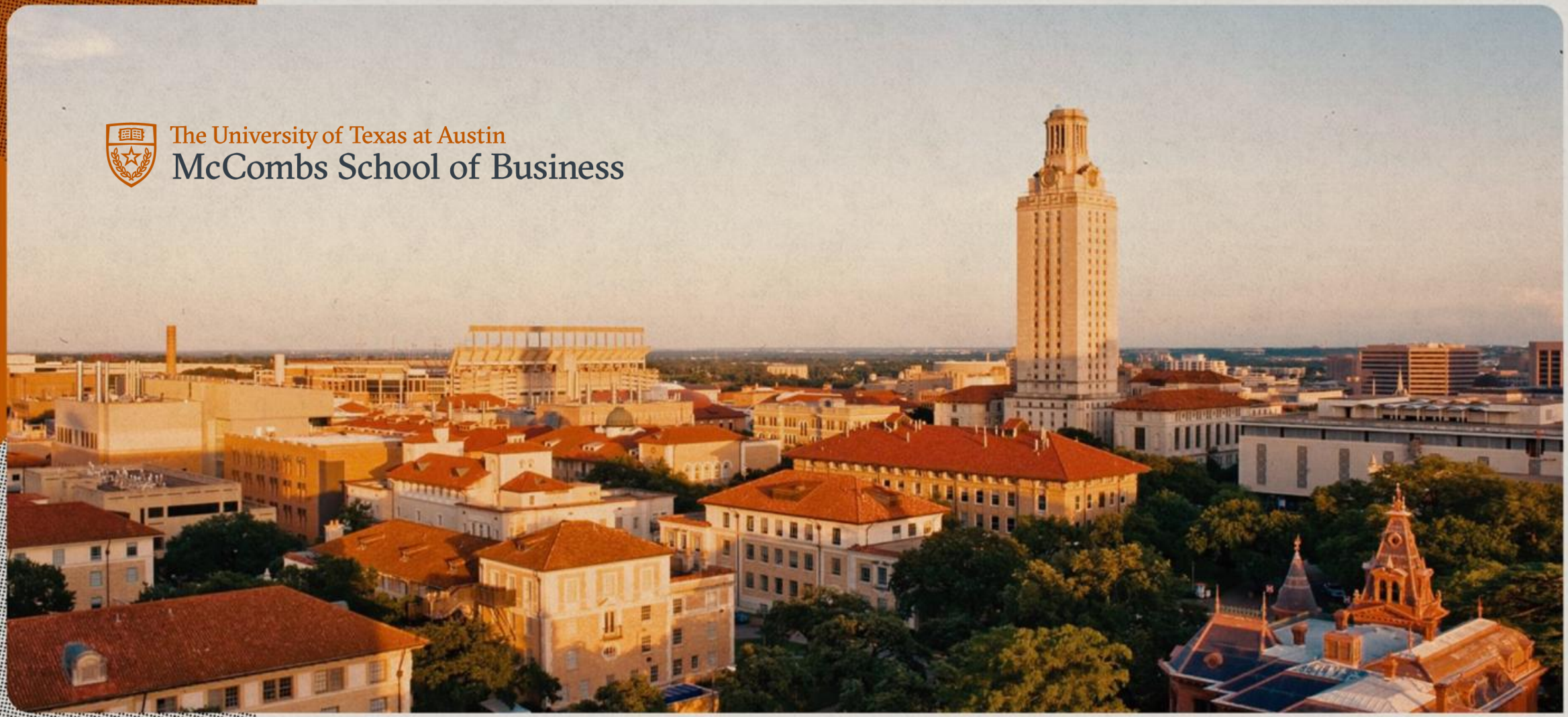

D3: *K*-means CA (Iterative Partitioning)

Learning Objectives

1. Understand *K*-means approach to clustering
2. Implement *K*-means in R
3. Understand need for scaling

Major R commands

kmeans

pairs, plot3d

scale

set.seed

kmeans sub-objects: $cluster, $centers, $tot.withinss,

$betweenss, $size

Agenda

1. *K*-means demo with *Stock data.xlsx*
2. *K*-means demo with *AustinApartmentRent.xls*
3. Your turn

Cluster Analysis from 40,000 feet

There are two major categories of CA:

- iterative partitioning (Day 3)
- hierarchical agglomerative (Day 4)

CA groups similar rows of data together in groups called clusters. Thereby, CA simplifies the data since, for many purposes, similar data may be treated as though they are a single unit. E.g., advertisers divide consumers into market segments; politicians divide the electorate into voter blocs; etc.

CA from 40,000 feet - IP

The IP process begins by choosing an initial set (K) of cluster seeds, which can be prespecified, algorithmically determined, or randomly determined. Seeds are usually selected cases, but need not be. Often spread evenly through the data. Thereupon, the first crop of clusters is grown by assigning each case to its nearest seed, by some distance metric, usually Euclidean distance. The centroids of these clusters become new seeds.

The second crop of clusters is grown by re-assigning each case to the nearest new seed. The centroids of the new clusters become the still newer seeds.

CA from 40,000 feet - IP

Cases are reassigned yet again. Iteration continues until no case changes assignment.

Main reason for failure of K-means: Poorly chosen initial seeds.

To decide the number of clusters (K), must repeat whole process for a range of values of K. Then compare resultant clusterings by some criterion or criteria.

Comparison criteria include: Elbow method, Silhouette method, Gap statistic method, R-square, AIC.

CA from 40,000 feet – R-Square

Every clustering (regardless of method) has an “R-square” that measures the proportion of the data structure that the clustering explains, from 0 (one super-supercluster) to 1 (all singleton clusters).

The R-square of a given clustering = Between Sum of Squares ÷ Total Sum of Squares = 1 - Within Sum of Squares ÷ Total Sum of Squares = BSS / TSS.

BSS = data variability explained by the clustering.

WSS = data variability unexplained by the clustering

TSS = total variability of the data = BSS + WSS.

CA from 40,000 feet – R-Square (1)

The clustering R-square can be thought of in two equivalent ways:

(1) **The notion that a good clustering has small clusters widely separated.**

Measure size of clusters by total variability within clusters (WSS), and measure cluster separation by total distance between their centroids (BSS).

WSS = total squared distance from points in a cluster to its centroid, summed over all variables, summed over all clusters.

BSS = total squared distance from cluster centroid to overall data centroid, summed over all variables, summed over all clusters

What is the value of the WSS for the clustering that has each data point in its own singleton cluster?

0

Single choice poll

1

Single choice poll

can't say

Single choice poll

What is the value of the BSS for the clustering that has all data points in one super-cluster?

0

Single choice poll

1

Single choice poll

can't say

Single choice poll

What is the value of the BSS for the clustering that has each data point in its own singleton cluster?

0

Single choice poll

1

Single choice poll

can't say

Single choice poll

CA from 40,000 feet – R-Square (2)

(2) The notion that cluster membership indicators of a good clustering should well explain the variability of the data when the data are regressed on the indicators.

Create 0-1 indicator variable for each cluster: 1 = row is a member, 0 = row is not a member. Regress each clustering variable on all but one (to avoid multicollinearity) of the indicators. Get SS (Sum of Squares) Explained for each regression. Add them up. Will equal BSS. Get SS Unexplained for each regression. Add them up. Will equal WSS. Then R-square = BSS / (BSS+WSS).

When clustering people by salary and wealth, if you have two clusters (C1 and C2), you can calculate the R-square of the clustering by regressing *[pick best answer]*:

salary on C1

Single choice poll

salary and wealth on C1

Single choice poll

salary on C1 and C2

Single choice poll

salary and wealth on C1 and C2

Single choice poll

CA from 40,000 feet – Interpretation

The interpretation of the meaning of a cluster derives from the values of the clustering variables that characterize the cluster's members.

Suggestion: Calculate the means of all variables by cluster. Weave a story that links the values of the means consistently and distinguishes the story for one cluster from the stories for other clusters. Story-telling is the most important practical skill in cluster analysis and is as much art as science.

K-means Demo

CA demo (kmeans).R

Uses *Stock data.xlsx* and *AustinApartmentRent.xls*

Illustrates use of *kmeans* in R and need for scaling.

Your turn

Continue cluster analysis of *AustinApartmentRent.xls* by using *kmeans* to find and interpret 6-cluster solution using all variables except Rent.

How many apartments are in each cluster?

What is the R-square for the 6-cluster solution?

How to interpret each cluster?

Is the solution stable if the clustering is repeated?